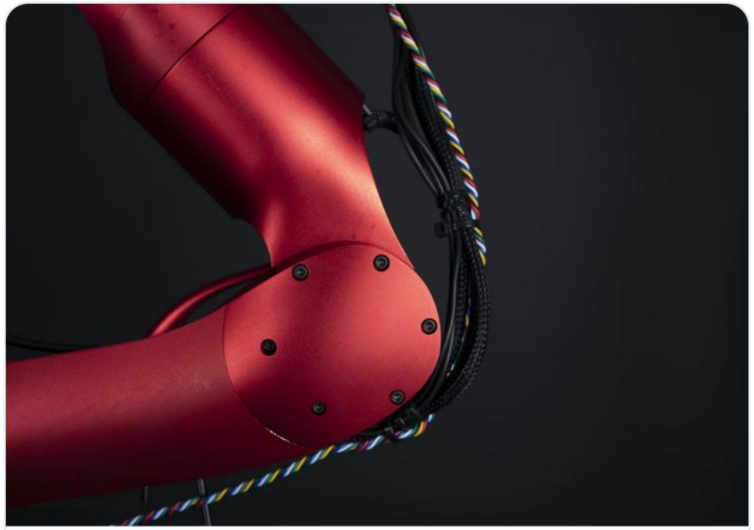
ROBOTICS ENGINEERING STUDY

Types of Robotic Joints

Discussing Revolute Joints, Prismatic Joints, and RR Mechanisms with creation and movement principles.

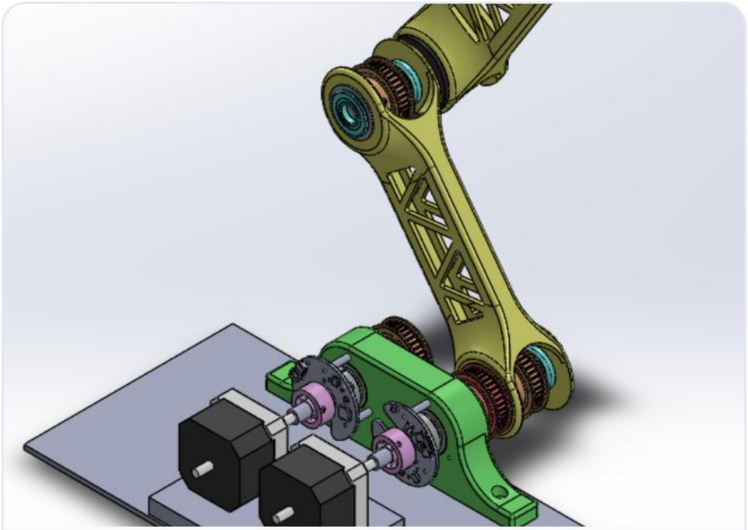
SUBTOPIC 01

Revolute Joint (Rotary Joints)



Axis Rotation

Allows links to rotate cleanly around an axis in different directions.



Robotic Arms & Legs

Commonly found in articulating robotic arms and walking robot legs.

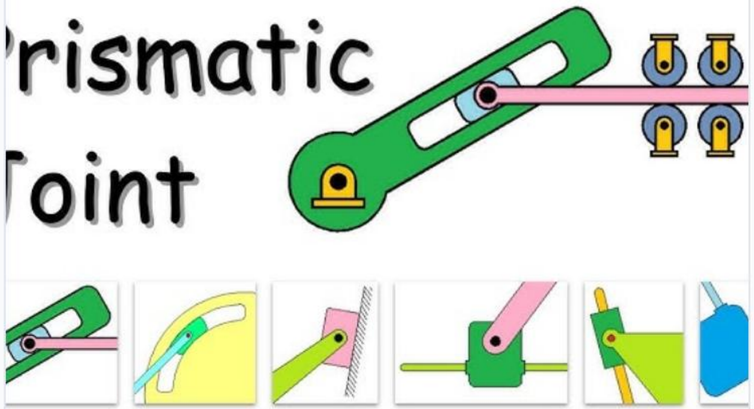


Tinkercad Visualisation

Created and visualized easily via 3D software models.

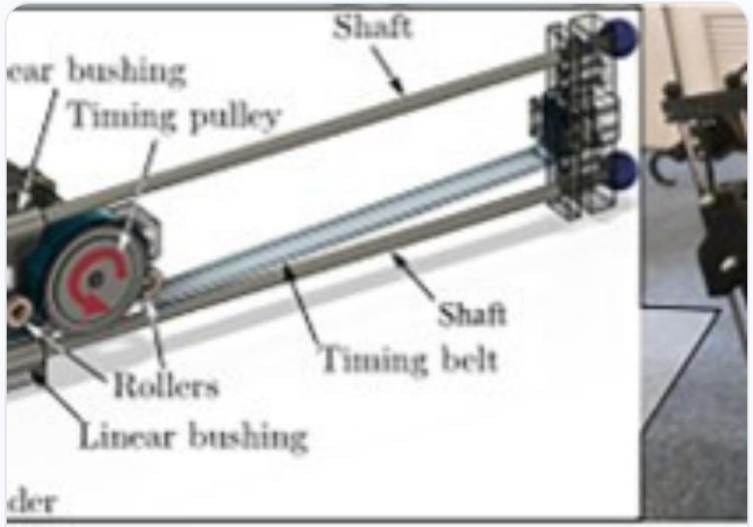
SUBTOPIC 02

Prismatic Joint (Linear Joints)



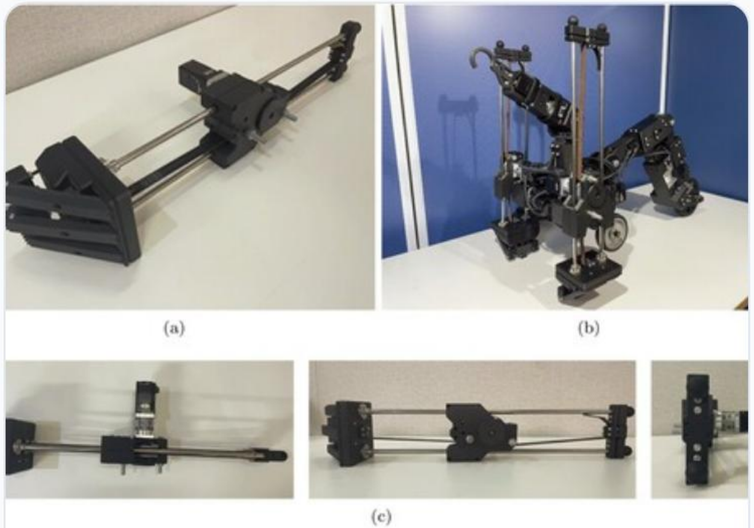
Linear Movement

Allows links to move in a straight line along an axis.



Common Example

Comparable to the sliding mechanism found in a drawer.

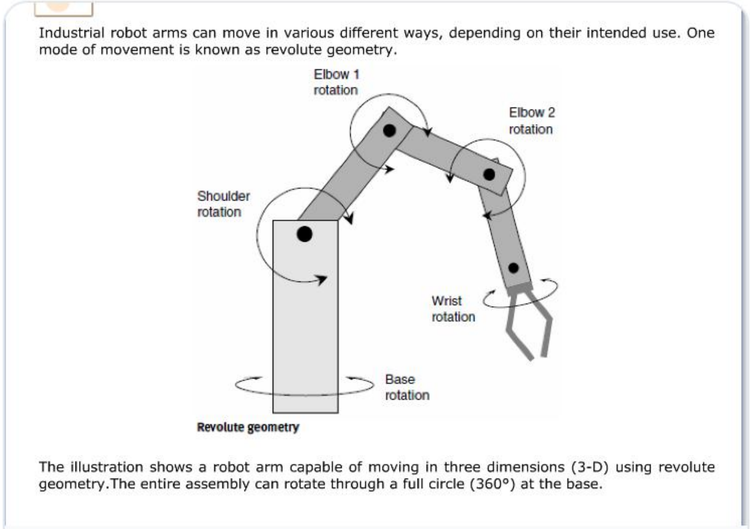


Activity 5.4 Design

Built and tested using Tinkercad modeling environments.

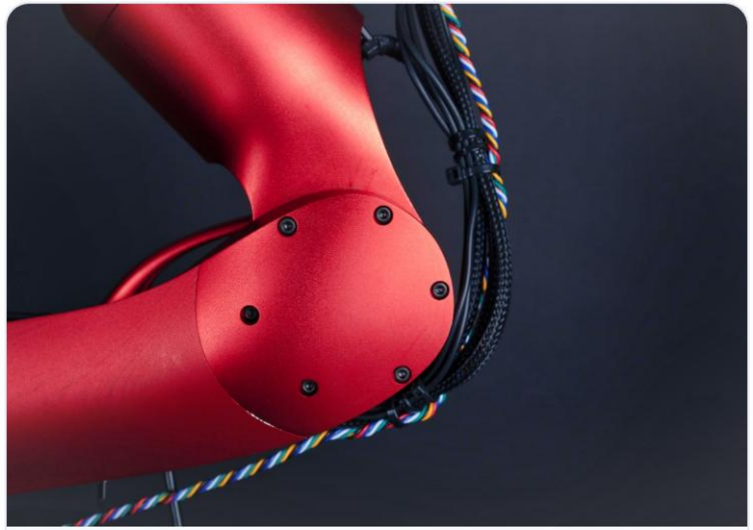
SUBTOPIC 03

RR Mechanism (Revolute-Revolute)



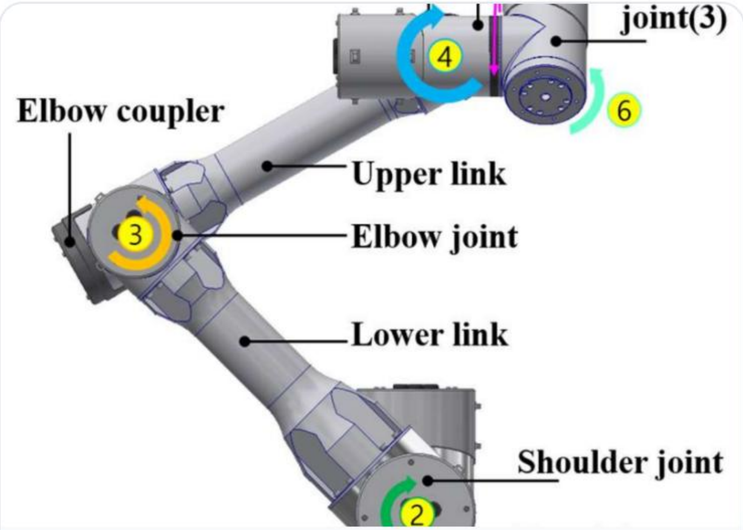
Two Revolute Joints

Involves the use of two revolute joints connected in series.



Robotic Arm Application

Most commonly applied in robotic arms for enhanced articulation.



Shoulder & Elbow

Joints placed respectively at the shoulder and elbow positions.

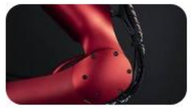
SUMMARY

Mastering Robotic Joint Mechanisms

Understanding Revolute, Prismatic, and RR mechanisms enables effective 3D design and kinematics in robotics.

Revolute Joint | Prismatic Joint | RR Mechanism

Image Sources



https://wevolver-project-images.s3.us-west-1.amazonaws.com/0.7gqrnx8t8zeimage5.png

Source: www.wevolver.com



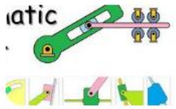
https://westsideelectronics.com/content/images/2019/01/image-11.png

Source: westsideelectronics.com



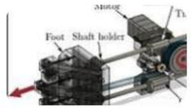
https://kincorobotics.com/wp-content/uploads/2026/07/Rotary-Joint-Module-for-Humanoid-Robots-iFMH60-main-800×800.jpg

Source: kincorobotics.com



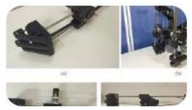
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Source: www.youtube.com



https://www.tandfonline.com/action/showGraphicalAbstractImage?doi=10.1080%2F01691864.2025.2571558&id=tadr_a_2571558_uf0001_oc.jpg

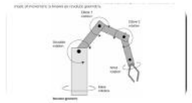
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Image Sources



http://www.robotpark.com/academy/EC/21116-REVOLUTE-GEOMETRY.png

Source: www.robotpark.com



https://robodk.com/blog/wp-content/uploads/2022/05/Parts-of-Robotic-Arm-Joints.jpg

Source: robodk.com



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Source: www.mdpi.com